

ActInf Livestream #034.0 ~ "The free energy principle: it's not about what it takes...."

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[Music] all right hello and welcome everyone it's active lab live stream number 34.0 on december 3rd 2021 and we're discussing the paper the free energy principle it's not about what it takes it's about what took you there welcome to the actin flap we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this slide this is a recorded and an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be following cool diatic video chat etiquette today at this coda site you can find our sortable searchable interactable and growing database of past and future live streams and if nothing else isn't this paper going to be about the past and the future and it's also our last paper discussion that we're kicking off for 2021 in number 34. so it's a pretty cool discussion and i'm really looking forward to it in actin stream number 34 we're going to be discussing the paper the free energy it's about what took you there by axel constant in 2021 and just like all the other dot zeros the video is a introduction to some of the ideas and the context for the paper it's not a review or an evaluation or a final word we'll go over the aims and the claims the abstract and the roadmap of the paper and then there's a couple keywords including some that we haven't seen before and then we'll walk through the key example which is the numerical example of a simple signal detecting bacterium and we'll kind of talk through that example and see where that takes us and i'll just note as with all the other dot zeros check out the paper if you want to learn more and axel's writing is really a pleasure to read and he has deep insight into the fep so we'll start with just saying hello and anything that we else wanted to say so i'm daniel i'm a researcher in california and what excited me about this paper and got me excited to discuss it was i think axel's provocative as well as grounded and respectful approach to dealing with what has and hasn't been done in the fep and the example is so minimal and paired down but also insightful that i think it lends itself to a jumping off point for many many other discussions dean yeah uh uh thanks daniel i'm dean i'm

i'm in calgary and uh i think what excited me about this was that i um i didn't start out really understanding what the title of the paper even meant but then when there was a bit of an inversion and a bit of a taking things inside out and i don't know if you want to describe that as a reversal of what is typically viewed as the time continuum or not um it just opened up a whole bunch of uh interesting ideas which i was contemplating as a as a program developer because i knew that um in order for the idea of being able to go from an inference i believed going from an inference which is a very small time frame through a prediction set and finally into being able to get to a place of modeling or understanding co-variation as a as collective uh this paper would probably help and then with each section i went through it was it was very helpful in terms of being able to sort of bring this idea just as an idea not in terms of its practical implications but just as an idea to life so okay i'm sure he'll rejoin i'm gonna just read through some of the big questions and again these are just the ones that we wrote down so all read the first two and then maybe say something and then dean you can take the second two so two of the big questions that the paper raises are what is the free energy principle about and what is the entailment relationship between life and the minimization of free energy or another way to read that is what is the entailment relationship between life and models that emphasize the minimization of free energy i just love how the paper dances between both of them and figuring out what the fep exactly is and isn't in its distinction from general bayesian theories from active inference these are really crucial distinctions yeah so the the second two questions were one one was from the author i'll read the fourth one first from the author are prior subjective or objective under the fep um i put the words and with the question mark yes because i actually have a question about this he he comes to a conclusion but i'm not i'm not sure what the timeline is switch leads back to bullet three which is what can we say about relationship when we shift from looking at what something is the comparison of life and free energy principle to how long it lasts he talks about duration or the when aspect of that and that's the thing that kind of adds a twist to this because you can look at it as very momentary or you can look at his things as very cyclic obviously expression episodic and diachronic and you've used kairos and there's lots and lots of ways to bring that in but when we do things change pretty quickly and it's so funny you had to go to the fourth one but was it really the fourth question if you read it third right there we go so the paper was published on two two 2 2 1 could have waited a year and gone for the total too but i understand you got to get those papers out and a few of the key aims and claims so first using the author's words the entailment problem is the confusion in the entailment relationship between free energy

minimization and life so if you've never heard the word entailment never heard free energy we're going to go into some of these things but the entailment problem is people being confused about entailment that's a problem in this paper the author dissolves what's called the entailment problem proposed here by providing a numerical example of free energy minimization in a hypothetical organism i conclude with some brief epistemological knowledge oriented remarks that may be of interest for those who worry about the explanatory power of the fep and then this is the last words of the paper which we'll return to at the end of this live stream the claim i set up at the start of the paper is that free energy minimization is not sufficient for life mic drop fidel true to the unifying grip of hypotheses in historical sciences the free energy principle is meant to account for all of that you me and the organism under study in a unifying fashion any thoughts on those claims we'll be exploring them and discussing them in the today yeah no the only thing that i immediately went to is was it was a question that i would ask people who were sort of on the other side of whatever it is that they were moving towards in terms of a sort of a professional participation participation and so the entailment thing for me was is it opportunity over method meaning method entailing opportunity or or is it met opportunity being over method or method being entailed by opportunity and that is the same kind of question that that axel looks at here so depending on how you see that entailment things go off in completely different directions so let's go and get into the paper now because yeah that's something that you can't resolve until post-diction you can expect how it'll be resolved but then you look back and it might have been different so i'll read the abstract and then feel free to give a thought and of course anyone watching along live is always welcome to give comments and questions in the live chat so the abstract is i'll read um the first and the third one philosophical writings on the free energy principle in the life sciences often give the impression that minimizing free energy is sufficient for life but minimizing free energy is not a sufficient condition for life in fact one can perfectly well conceive of a system that actively minimizes its free energy and for this very reason moves inexorably towards death where does the assumption of this entailment relation come from there is indeed an entailment relation but it goes the other way around life entails minimizing free energy but another way if you exist now under the right conditions it's because you've done something like minimizing your free energy however the question of whether you exist tomorrow cannot be settled purely by the resorting to the fact that you'll minimize your free energy in order to get there the simple point i make in this paper is that the free energy principle is not concerned with the sufficient conditions of existence but rather what must have

been the case necessary what must have been the case given that you exist it's not about figuring out what it takes to be alive it's about figuring out what took you there and i'd recommend people to put this in a speed text-to-speech reader it reads just so fantastically in a robotic voice i rarely do that for papers but i was on my exercise bike listening to the robot talk about this and it was just it was hilarious so here's the simple point in the paper free energy principle is not concerned with the sufficient conditions of existence but rather with what must have been the case given you exist it's worth restating because it's a simple point but the depth of how far off one can be by missing this point is vast it's not about figuring out what it takes to be alive it's about figuring out what took you there not what took the living system there so here's the living system but we can even finesse that it's you and so the red line is the system looking back how did i get here or putting kind of in a second person way looking back at another lifeline how did i get here the question that is often asked by living systems including us is is something alive that's answering schrodinger's question ramstead at all 2018. it's active topic of discussion and so here's the line of another system and so it's looking at the past the present and the future of some other system is the fep about this red arc of the system projecting back and asking what took me is it looking out to another system and asking whether it's going to be the ruler of like the meter stick of what is alive or not so probably other ways to represent it any thoughts on that dean yeah it's just what is alive thing now can be placed beside and when is alive and i think that's what this paper really brings forward if we look only at what is alive we may over reduce may but if we include when is alive as you said things get real interesting really really fast and a really influential paper was fristen's single author 2013 life as we know it and so that's often seen to be making a claim about the blue because it's like real life as we know it this is like how we know what's alive but then it's like even in the title with so few words as we know it right so even in there there's a little bit more depth so this is a nice road trip there's only a few stops few sections and figures so there's an introduction and then minimizing free energy for better or worse basically just like you could minimize your potential energy right off a cliff you can minimize free energy right off a cliff for better or for worse assuming life is better we prefer to be alive there's a section on the conceptual distinctions between bayes and the free energy principle which is very didactic there's a numerical example presented in formalism as well as in a figure which we'll talk about a lot today and then free energy on a wing and a prior the last section is called future direction free energy minimization as a historical scientific principle which i thought was exciting and appropriate for the paper being in biology and philosophy so

for those who are not steeped deeply in the philosophy game there's some different terminology but that's why the whole field is enriched by people bringing these different perspectives to bear so the key words that are brought up prediction and post addition before and after historical sciences the free energy principle and then i think we added active inference and strong and weak claims they're argued they were given as a keyword but they are key words for this discussion so let's start with prediction okay i'll just say what i have on the slide then dean our resident prediction matter expert so um first never make predictions especially about the future casey stengel or as they say in trading either predict the price or the date but never both and on the wikipedia page for prediction there's so many different related areas and that just tells you like well what is it that's related to all these areas things that are spooky and forward thinking like an omen or an oracle like almost metaphysical or supernatural all the way to things that are very technical reference class forecasting regression trends expectations forecasting and i think in active there's multiple ways that prediction comes into play so just listing only four of a certain set there's the prediction or the now casting of sensations and that's like predictive processing like the top down priors are modeled as providing predictions about sensory inputs there's the prediction of or inference of hidden causes in the world from those sensory inputs which are not directly observed there's uh and importantly notice that these two predictions are actually almost instantaneous and then we also talk about prediction through time like at time t there's the prediction or the anticipation of states at a future time and those could be internal external sense action states so predicted future states and then there's also the prediction of the consequences of action which is anticipatory cybernetics and can that that can include um counterfactuals so what if something were to happen that's like a prediction about a counterfactual so there's multiple ways that prediction and active inference come into play what do you think about that dean yeah i think it's all true i think it just points if we're if we're going to ask how time is included then we can break out how time has been included like what the difference is in terms of how time is included in an expectation versus how times you is included in a trend estimation because i think that's what causes us to be able to see multiple different ways of prediction so not just types of prediction but then the actual time given over to that particular type great those who study prediction might be considered um future studiers or futurologists and that's also an interesting wiki page and search area just like how many frameworks were brought up i mean they forgot one it starts with active and ends with but there's a lot of different methods here and i'm sure we could all think of more but it's kind of like we have prediction as a concept

looking forward in time or even to the now which is kind of forward in time from when we made that prediction and then there's the field around that because we're going to also look at and then the fields that deal with so just whether you're hearing about active inference for the first time or not think about what is prediction fundamentally looking forward in time and then what are the attributes of fields that look forward in time for example they may not have proof because they're making a prediction about something that hasn't yet played so they might be an estimate that you'll bet the house on but can one say that they have proof when they're talking about something going forward in time now now this is where it gets really interesting because if you're an archaeologist slash anthropologist and you're trying to figure out whether the plains of salivary where salisbury was the originating place for all those rocks to get tipped up you're actually looking forward into a future that your you think is maybe unresolved even though if you're looking into the deep past as well so again i know that time time 0 seems to be going only one direction but our minds don't seem to necessarily honor that exactly so we can contrast prediction with post-diction and so quoting the paper as a reasoning pattern the fep can be used to generate postdictive statements in the sense of first in it all 2017 which we'll talk about accordingly fep reasoning might be interpreted as a principle akin to those found in the post-dictive sciences aka historical sciences hashtag keyword like geology paleontology archaeology or any science that deals with irreproducible causes cleland postdictive scientific statements are concerned with what must have been the case instead of what will be the case cleland's paper from 2002 is posted here it's very clear to see how the paper uh under discussion draws on this one and just to read the conclusion of the abstract i show that these different patterns of reasoning are grounded in an objective and remarkably pervasive time asymmetry of nature so there is this like entropies arrow times arrow and that makes talking about the future and the now different than talking about the past that's the historical sciences and then the first general 2017 paper now this is interesting because postdictive was only used one time in the paper figure 2 illustrates the recurrent nature of the message passing that mediates this predictive and postdictive inference using little arrows one can clearly see that the first outcome can influence expectation about the final hidden state and expectations about the final hidden state reach back and influence expectations about the initial step this will be this will become an important aspect of the deep temporal models considered later so here's that figure too it's the partially observable markov decision process that we know and love with a layer of the message passing added because um par and devry are authors and this is like a very i mean 2017 wasn't so long ago but this was an important

work that combined a lot of different areas together and here's that reach around through time in an interesting way yeah this is where bringing your history and my history into this interaction creates a new history and i know that's symbolically i'm not made that's a gross over simplification but if you were trying to share this with somebody with not as much background maybe that would be a simple simple tell i don't know cool and just like we talked about future studies let's think about the historical sciences so here we have the past present future and historical sciences can refer to history proper the auxiliary sciences of history or any science that draws from records of past events aren't experiments also past events now the auxiliary sciences is include but are not limited to and there's so many fun fields here things that are like kind of like subfields of history like heraldry paleography so there's some great you know vexillology a lot of awesome terms in here and then i was thinking well you know you can't prove something in the future and it hasn't happened yet and then historically you also can't prove it because it couldn't have been otherwise hence you have to only say what must have been the case instead of what will be the case like you can't generalize you can't say well the galapagos are biodiverse because and then just give one singular answer even putting 10 virgin aside and everything then so what's left the past the present and the future and then the now is just this like kind of moving indicator that we're experiencing and so it doesn't really make sense to prove something only in the now either so that's just very interesting how there's that as clelland calls it the pervasive time asymmetry of nature and you know put your hand down if you disagree what does that leave for us to actually work with in terms of scientific modeling and from that tension is actually where fep and active are going to be doing some interesting things yeah this was if you're this is what i love your the the images that you put up here because one of the things from from a learning standpoint that i found really helpful is instead of a test being what answers can you can you pull from your memory bank it was a learning exhibition where the chronology was kind of the slow hand you'd lived x number of months of working in a professional setting and in front of a significant audience you were laying out a chronological timeline where the focus was on this the feelings now now top down organized and narrated but that whole that entire picture of feelings of working in a professional setting for months actually now came out in terms of a narrative so the historical sciences piece here is actually an amazing consolidator as well and that's why that's what i found about this paper it was actually an incredible consolidator agreed so necessity and sufficiency there are different words they mean different things and sometimes it's easy to get it right in a simple case but then when science gonna science it gets like really mixed up

necessity means that you need it like the car needs wheels to roll normally if you take it away it's a loss of function the wheels are a necessity for the car but are wheels a sufficiency for the car well no because just four wheels doesn't equal a car or you could have all the car but if there's no gas like there's something else that's also necessary they're very similar terms they both describe the conditions that are sufficient and or necessary for something being the case and that's been described discussed in the math threema in this figure in the mathematical case of course necessity insufficiency is a key topic in the gene regulatory network case that i've studied where they talk about genes being necessary or sufficient for a given phenotype and then also in a legal case from this yale law journal talking about the most restrictive to the least restrictive like if somebody had no causal role in a crime that inarguably occurred they're blameless if somebody was necessary and sufficient for something to be caused it makes sense that they would be guilty of that being occurred but then there's complex systems in between and there's the difference that makes a difference and there's so many ways to talk about necessity and sufficiency so there's sort of a simple bookmark on both necessity and sufficiency and neither and then there's a complex space in between all i would say is no matter how often you talk about causation it's amazing how much stuff just gets roman candle information wise out of that simple word that's all i need to say about that the last key word proper is free energy principle and check out active stream number 14 number 32 number 26 i mean roll the die we talk about it a lot but it's always important to go over it because it may be someone's first time hearing it and the simple aspects of it are the parts that we want to return to again and again kind of like a strange attractor in our model kind of like the fep we've talked about some of the technical part on the bottom so check out especially 32 and 26 for the bottom parts and then the free energy principle is this imperative to be self-evidencing and anti-dissipative if i could only choose two words potentially those would be it and they're both going to come up today because anti-dissipative will be equated to persistent duration of survival and then evidencing is going to be related to bayesian approaches to modeling evidence and then where do you go from there there's so many places to go active inference the corollary process theory and um we're going to talk and unpack fep in a few different domains so good to move to the next slide all right so how does fep free energy principle relate to life in this paper the paper says minimizing free energy is the process whereby one maintains one's structural integrity in phase of environmental perturbations by revisiting one's most probable organization of physiologic states it is in that sense that minimizing free energy is considered a condition for one can equate survival with life since one supposes the other

under the fep and then kind of two slogans cool sentence structure if i survived it means that i maintained my structural integrity in face of environmental perturbations in a dissipative universe according to the era of time and maintaining my structural integrity is what qualifies me as living from the point of view of the fep when considering metamorphic organisms it may be said that this the life cycle that corresponds to the thing whose integrity is maintained over time eg over evolutionary time not the specific form that the system takes at one stage of its development e.g the adult form of a frog or ant this casts the fep within the realm of process ontology versus substance ontology and i just have to say that bucky fuller talks about structural integrity as being like the house that stays up through time pattern integrity is in his ontology a system that recapitulates itself and so it retains dynamical integrity and um then that's connected to the idea of process ontology versus substance ontology which we'll discuss more on the next slide but it's just awesome like this is an ants an ant has a life cycle but then you can almost pull back and say well really the integrity of this egg becoming an adult is related to the nurses feeding that egg so even if that egg to adult transition reminds us of a solitary organism the integrity that's being regenerated generation after generation is like the colony and so what are we talking about the persistence of when you never step into the same river twice like in one sense there is no integrity being maintained because things are always changing and so the fep framing life in that way is very interesting what do you think yeah this is where it's i would often say to people i want you to go into this situation i know it's new for you but i want you to go into this situation with both eyes open and with one of the eyes i want you to see the project at hand potentially and with the other i want you to be able to see the system at hand potentially that's very confusing because i mean the the best that we most of us can do is just deal with the complexity of the project or the complexity of the system not the complexity of both co currently but get having said that the better you can work at knowing both are present regardless of what you decide is your focal point in that particular moment the better you'll have a an opportunity to sort of get the whole picture going on here it's hard in fact it takes it takes some moving away from the way that most people are trained to try to look at things so it's really hard to be honest let's dig into process versus substance ontology perfect so the 2020 dupree paper is cited life as process and also just a cool kind of coincidence that it's a bilingual abstract if not paper with english and russian check out the paper and all the citations if you want to read more but just to read the highlighted parts that kind of get the essence of the abstract the metaphysics that has dominated western philosophy and that currently shapes most understanding of life and the life sciences sees the world as

composed of things and their properties while these things appear to undergo all kinds of changes like metabolism development context changes it has often been supposed that this amounts to no more than a change in the spatial relations of their unchanging from antiquity however there has been a rival to this view the process ontology associated in antiquity with the fragmentary surviving writings of heraclitus for process ontology what most fundamentally exists is change or process what we are tempted to think of as constant things are in reality merely temporary stabilities in this constant flux of change eddie's in the flux of my main claim in this paper will be that a metaphysics of the latter kind is the only kind adequate to making sense of the living world and also related research is this causally powerful processes from earlier in 2021 by dupreeh in synthese so this is a fascinating line of research about moving from a substance to a process-based view and where causation comes into play with with respect to the historical sciences there's so many other ways to take this another quote that one of my philosophy professors would always say and i think it was a quote from kant was there will never be a newton for a blade of grass because there isn't a mechanistic explanation that's timeless for a blade of grass in the way that there plausibly could be for like newton's cradle so i wanted to ask you did you find this have you known about this before reading this paper or did the paper um cause you to go and look and find papers on process ontology because two of our colleagues talk about flux blue and and stephen a lot and so when you brought this forward i thought okay so has daniel been hiding this from me all along because he's kind of known about this network or did the paper in effect create a situation where you wanted to know more about process ontology i followed the stigmergy in the paper it says it's unclear whether um it might be argued life and survival refer to an enduring process that advocates of process oncology would call organism in the sense of depre 2020 went down to the citations there was yeah i knew this philosopher i knew this philosopher but i hadn't read this paper yeah so exactly process entology now generating new process ontologies variation upon variation right and still still at that fep narrow time frame it's it's that's goes back to as i said i'm not sure if it's um just subjective or subjective and objective so anyway i'm glad you found this because now i got to read dupree because yeah anyway oh substance where are they sting how does the fpp relate to active we're kind of getting ahead in the paper in one sense so we'll be coming back around to it but the fep on its own is a principle namely a foundation for reasoning about things eg living things in this paper we approach the fep as such however the fep can also be read more broadly as a research program that uses fep reasoning patterns to generate scientific hypotheses this involves implementing fep reasoning

into a theory called active inference which is routinely used to study various cognitive functions when you go to the footnote there's only one it's important to note that fep includes processes other than free energy minimization it also includes expected free energy minimization and generalized free energy while minimizing free energy endows the organism with post-dictive inference which we're gonna go into in the example and talk about more minimizing expected free energy endows the organism with predictive inference fap is postdictive it looks at the oncoming stream of data and it compares that to basically what must have been the case but active or any other predictive theory that's about expectations it kind of has to be different because the consequences of action are in the so it has to be these two different modes we're gonna come back to these ideas but fep reasoning yields historical hypotheses and not just like epic or legendary but they're about the past because it operates a historical evidentiary pattern and provides a compelling unifying causal story it operates a curious evidentiary pattern though because it assumes that both the investigator and the thing under investigation conform to that that pattern is free energy minimization per se so if you're making fewer prediction errors you are your your time frame carries on but you actually do become i don't know if you want the label but you do become somebody more versed and post-dictive what do we want to call them post-dictive um confirmations i don't even know what how to describe it but this was a huge thing for me post-dictator i guess yeah we talk a lot about prediction markets but where's the post-diction market and what if organisms are so good at post-diction it looks predictive so let's see yeah it's going to be you know they've got disclaimers whenever you read the forward-looking statements right like this it postdictive is now a new term that i can i can pull into those conversations around the disclaimer it's like if something's not predictive it's either about right now or it's post-active let's get to the core terminology that axel is bringing up here and like this is what reading the paper means is to understand like why this entailment problem is framed what responses exist to that problem implicitly explicitly how it's addressed in this paper and what the implications are going forward that's like you'll have read the paper to have known those things this was an earlier quote that sometimes arguments in the literature in the fep give the impression that in order to be alive eg sufficiency one must minimize free energy such a claim does not straightforwardly apply to the fep however and this is what the paper will demonstrate minimizing free energy does not entail candles minimize fe too rather the argument is that if you're alive it probably means that you have done something like minimizing your free energy which is the bay's optimal thing to do when your life depends upon solving complex inference problems this is a

subtle but crucial point to getting the story straight i shall call this the entailment problem that is the confusion in the entailment relation between first between free energy minimization and life here the notion of entailment refers to the implication i.e first order logical property between free energy minimization and the fact of displaying some life-related processes let's think about two different islands minimizing free energy and life so we could think of it as like a venn diagram they could be overlapping they could be partially overlapping they could be non-overlapping arrows can be drawn anywhere as long as you label them actual frames there to be two different kinds of threads like two archetypes of framings in the literature the first is the strong claim the strong claim is that or what's called the overly generous claim by kirschhoff and froze 2017 minimizing free energy is a sufficient condition for life so we're going to go out there with our free energy thermometer and things that are minimizing free energy are alive because it's sufficient the other claim is the weak claim if a system is currently alive it means that it minimized its free energy in other words minimizing free energy was necessary but not necessarily sufficient for being alive such a claim does not assume the fap is designed to set the bar for the sufficient conditions of life or meant to predict what things may or may not be alive rather it limits the scope of application of the principle to beings that we think are alive now what made us think that that's not in the theory and enables us to know the necessary conditions under which those beings can be living i.e can actively resist the loss of structural integrity what took them there what do you think about that well i just think it's interesting that he talks later on we'll get to this is that that you need the correct or right priors and and if you go with a strong argument you assume that the right priors going forward will continue to just materialize versus the weak argument which says it's not until afterwards whether we know whether or not the right priors were the ones picked there's so many fun examples so let's talk about bayesian organisms bayesian approaches to animal behavior whether it's bayesian approaches to foraging or the bayesian brain proposed that one can model red light one can model not that organisms are right one can model so just to forestall any kind of map territory distinction which we always return to but can't be returned to enough organisms as representing their relation to environmental states using priors and a likelihood let's call those representations bayesian beliefs not in the organism it's our representation of that in the model on the basis of those beliefs organisms generate adaptive behavior bayes beliefs can represent the probability of environmental states prior to observing a signal that's called a prior means before the relation between environmental states and the observed environmental signals aka a likelihood bayes theorem from which terms such as the prior and

likelihood come from is typically expressed as an evidentiary relationship between some prior hypotheses $p(h)$ and the observation or hand or data that's e so e is going to be like evidence or empirical h is for like hypothesis and the way that this sentence is read this expression this is bayes theorem and it's read as the probability of the hypothesis given the evidence vertical bar means given or conditioned on so the probability of the hypothesis given the evidence equals there's the equal sign both sides are the same the probability of the evidence given the hypothesis multiplied by the probability of the hypothesis that's in the brackets divided by the probability of the evidence bayes theorem people use bayes theorem to model all kinds of things about different systems the free energy principle is a bayesian formulation of the manner in which organisms infer the posterior probability of their prior beliefs after having a verve observed an environmental signal with a given likelihood and in doing so infer some hidden or unobserved variables same equation as the last slide what stands for the h are unobserved variables whose prior probability forms the hypothesis and what stands for e are the sensory signals organisms receive the data so this is like i think it's equally likely that it's day or night like it's 12 and 12. so 50 50 on whether the latent cause of the world the hypothesis is daytime or night time but you can't observe daytime or night time all you get are photons on the retina so then it's like you have a prior that's like 50 50 and then evidence comes in photons and then if there's a lot of photons you update your prior to get the posterior and then you think okay it's daytime because i'm getting a lot of photons hence it is often said that under the free energy principle organisms are viewed as embodying hypothesis belief or best guess about the cause of their sensations or sensory signals that they receive that's because the evidence for an organism are its onboard sensors and so the hypotheses are about causes of what is giving rise to sensation just like in the daylight example it was a hypothesis about what was giving rise to the photons because beliefs are embodied by the embedded enacted etc and thus the organism's own states the uncertainty and the likelihood in the prior can be viewed as representing uncertainty inherent to the biological apparatus instead of uncertainty in the world as would be the case under typical bayesian models under the fep uncertainty should be thus read as reporting a bayesian credence or credibility score over the organism's own beliefs as it reports the probability of a state or hypothesis relative to other possible hypotheses what do you think dean well i'm gonna i'm gonna what do i think i'm gonna ask you a question back so is is this another way of saying an account of uncertainty right like so that that that that sounds like wait a second how how can we have an account of something that we don't know but is that what he's i know i don't want to interrupt the math

because i really like the the way he step by step goes through this but i when i read this again when i read this the first time i went oh okay well this is an account of uncertainty wait a second how can that be so yes uncertainty is about um the hypothesis giving rise to so the way so so the when it takes place over a duration because the updating of priors is happening so it's like as my tummy is sending me signals the body is as if determining whether i'm hungry or not um uncertainty is the probability of a state or hypothesis relative to other hypotheses yeah so the traditional models are methods that we use to try to account for what has been learned doesn't necessarily take this into consideration when in fact this is the basis for moving forward and for learning i think there'd be some interesting 1.1.2 discussions about uncertainty in frequentism versus bayesian probability and about what it means for a frequentist p-value-driven understanding of certainty or not i fail to reject hunger at the 0.05 p-value versus the bayesian credible interval or the base factor of being hungry is this or that relative to this conditioned on the sensory inputs literally hypothesis hungry or not conditioned on the inputs yes yeah anyway just it was again i don't want to take this off an intention let's get back to the math because he does a brilliant job of clearing this up this is such a cool discussion the simple answer to the question of whether priors are objective or subjective under the fep is that they're subjective where's my gavel it's it's over as we said they track the confidence over a system's belief priors do not conform to irrationality beyond the rationality of the inference per se but nor are they rationally unconstrained this always makes me think of like sort of the new atheist like if you were rational you would believe like we believe and then that kind of absolutist rationality making way for a multi-perspective rationality like i guess different people with different priors and sensory inputs implement their own local rationality and they come to different conclusions fancy that yeah there's a rationality beyond bayesian rationality comes from the way variational base is realized by the system and we're going to talk about that later it basically embeds when into that process the approximation to the posterior in the variational bayesian inference q is the prior used in the next cycle of the rational constraint over priors is the fact that the approximate subjective posterior cue will not only be bayesian but will also always be the best guess relative to what the true posterior ought to be in short under the fep even though priors refer to psychological states of the system or internal states of the system updates of the system make those priors an approximation of what they should have been had the prior been updated with exact base thus it might be said that priors under the fep cut across the objective subjective dichotomy they are subjective just like the gavel said above while satisfying a rational constraint mandated by the existence of the system yeah

this is the precise approximation argument that comes up often meaning it's not a question of do we need both we do but how do how do we have both available to us well we've got to keep both eyes open we've got to look at both the system and the project which is really hard because it's much easier to just one or the other and give our attention over to that thing that's why post-diction is a little bit easier maybe hindsight isn't 20/20. as we'll prove in this paper but it's still better than than making that making that bet with with absolutely no priors we talked about substance versus process right and then we have this subjective objective so it's almost like substance ontology maps with objective because it's like the object it's the substance yeah the what and then process ontology maps to subjective because subjective is relational and dynamic it entails something different the win yeah and then we can kind of lean more on one side or the other but really it's like they're nested within each other yeah we can't let if we let if we drop one or the other or if you hear people dropping one or the other you kind of have to bring them back to no they're both here they're both present and they're both available regardless of what you choose to make your focal point that's that's when they go well that's like just your opinion man right exactly yep so let's go to the crux of the paper which is the numerical example so um there's a there's some other pieces you know we're giving a linear ordering and kind of jumping through the paper we're going to think about a organism like a bacterium that's going to infer whether a or b occurs externally for the organism a and b are part of the class r and the representations by the organism are a or b so r can kind of be read like receptor states a and b are inferred when receiving a chemical signal part of the class s like for signal which can be alpha or beta so s is the signal molecule alpha or beta we have we assume that before observing any signal so prior the probability of a is p and the probability of b is one minus p so they're mutually exclusive and they form a probability together then the numbers given for a and b the prior and then the conditional probabilities this is where life and death enters the picture you got to guess correctly if it's um a if alpha is out there and you guess a you live if alpha's out there and you guess wrong you die so you want to be like on the diagonals you want a and alpha to go together and b and beta but you don't want the off diagonals you'll die let's look at it graphically in figure 1. so the right panel is a visual representation of the organism inferring its prior beliefs about the cause of the observations it makes and then it restates um several of these pieces so we have like the um prior and then we have the signals a and b out there um here's something really important for all you active heads out there there's no action involved in our we're not talking about active inference in more biologically realistic descriptions of behavior like chance at all 2020 which is where this

example is a simplified version of we need to discuss active inference in active inference behavior is the result of a different inference process that of an action policy this involves more priors namely about the transition between hidden states and often about preferred sensory outcomes so that would entail a model where it's like okay there's a probability of transition between alpha and beta out there and then there's gonna be a preference for alpha and then i'm doing action selection on tumble or not based upon whether things are going well for me or not so that's several more pieces to the model right but it's a toy toy example that's going to make the point that minimizing free energy is not sufficient for being alive but it is necessary and so we're just distilling to the simplest possible sensory processing example that doesn't even have action so maybe someone could argue that well adding action somehow makes this example null and void or whatever but i don't think that's the and so um this does involve more priors and action is then distinguished from inferring hidden states it's about inferring another hidden variable which is policy so the first line is the generative model the joint distribution which is multiplying the prior and the likelihood the second and the third line these two represent the prior and the likelihood so here's the prior it's 80 percent a priori that it's um that i perceive a and 20 percent a priori i perceive b and then there's this mapping with a and alpha and there's like more density there and then b and beta more density there and the off diagonals are like less the fourth line is a possible bayesian algorithm that could be used and then the fifth line is a variational free energy minimization which we'll talk more about but that's kind of this example is we're just thinking about the sensory processing of a bacteria or cell that is needing to correctly guess the cause outside when it's only getting a receptor on its membrane for example what do you think daniel do you think people will know what the difference is between the process that you and i and steven walk through in 33.2 and what we're looking at right here will they will they know just how different those two representations are what do you think it's a good question we can re-state that this is not an active inference model right it's also not even trying to be the most awesome bayesian bacteria model ever right this is like just using bayes theorem right to show that simply is not gonna cut it for being alive under this contrived setting but um when we were in 32 or yeah 33.2 yeah and we walked through the the matrices where there was a preference and there was the hidden states it was quite different yeah so check it out do you see do you see co-variation happening at all in this co-variation um one example of co-variation is like you need to believe that you have accurate sensors if you there needs to be a covariation between a and alpha if you don't have them and you don't assume that you have them are you not dead eventually or

pretty quickly that's sort of the um gentle invisible hand of selections sweeping those kinds of systems off the table yeah yeah okay and i'm just asking because when i saw this i the first thing i said to myself is my goodness this if there was a continuum between 33 and figure 1 of 33 or 34 0 man this would these two things are really really far apart they are um well you know let's let's go to 33.2 so in 33 2 this is like looking um here's what we had done just like last week or whatever earlier this week yeah um the observations are getting mapped to this prediction about internal about hidden states so that's actually this a matrix is also the mapping matrix maybe it could have been a little bit clearer because also a is like the hidden state that's being inferred but um this bacterium is just reflecting this part right here so the prior comes in it's like just this sort of right angled neighborhood where you have the prior beliefs and then we're going to have observations coming in and then a mapping to the hidden states what active inference brings in is this whole tree where we're going to influence through policy selection how hidden states change through time and that entails the preferences free energy minimization and the affordances so this is like just a sub component of active right all right we're going to assume that the computation is conformant to bayes theorem and that the posterior probability after observing the signal is going to basically update the prior towards what is most likely given what was observed we're going to infer the posterior probability of a after observing alpha so the prior on a was 0.8 and then the the probability of alpha observing alpha contingent on a was 0.7 so if it's a in reality then 70 percent of the time you you hear about alpha that's what is actually on the receptor and then the b is just one minus such and so seven point seven point three bayes rule is applied and you know you carry through the math and basically after you've observed alpha your probability of estimating that the external state was a went from point eight to point nine zero three two so it didn't take you to a hundred but it bumped you up that is bayes updating so that's a bayesian bacterium updating its belief about a hidden state of the world based upon a molecule that actually only has a 70 30 accuracy right and that's really important to like it sounds like 70 30 it sounds like that should almost like degrade your accuracy but it actually even if it were 51.49 it would still upgrade your accuracy a tiny tiny tiny bit now if it were 1 0 0 1 well then yeah observing it if it's perfectly associated is going to update your belief a lot but even something that's relatively noisy like 70 30 can still make you go from 80 to 90 percent sure this is just basically taking whether or not you've got a decontextualized instead of information versus a contextualized one and what the benefits of adding the contextualization does i think again maybe i'm over oversimplifying but i think that okay so with exact bayesian inference this could be computed

but potentially that's not tractable in biological systems especially if you start to think about how there could be like 500 metabolites um and that's called the black box problem or the solipsism or seclusion problem variously in the fep literature so how does the fep get around that in order to bypass the problem the fep models that inference as a proximate approximate bayes bypasses the direct evaluation of the likelihood that makes it a lot more tractable the central claim of the fep interesting to read is that changes leading to behavioral and neurophysiological responses in living systems conform to a form of approximate bayesian inference known as variational base so check out active number 26 to learn more about variational bays which basically relates to we're going to control the family of equations and then we're going to do model fitting on a family of equations and that'll be easy like we're going to constrain our equation to y equals mx plus b linear model well then we can fit it with a least squares error it's going to be a convex optimization we can do it these are different types of problems but similarly we're going to have a certain family of equations that we parameterize and that's variational base what citations did you add all i put in here was some stuff that i've been looking at like over a decade ago now and some conversations that i've had with people like morris moskovitch and reyna around the idea that neurobiologically is there some sort of an analog between walking an elephant walking across the savanna once and not really seeing the desire line if a trace is going if traces are going through multiple times how does that affect potential gene folding in the neurons and stuff like that and so it doesn't really it doesn't really i don't i don't think it builds on the idea of what you're talking about in terms of how we can how we might want to be able to set up software recognition software but i but over a decade ago um the idea that how something is grooved in eventually an approximation becomes something clearer and the energy that you have to put into that and that's where you had the one the above the top cat being maybe enter too energy intensive i just think it's really interesting because back over a decade ago fmris were supposed to be the thing that was going to give us a clearer picture and there were people even back then they were saying wait a second we want to be able to talk about when a really blurry picture becomes a more clear picture as opposed to this is what a bump in in in energy tells us they kind of tried we're trying to kill that idea more than a decade ago in that example everything went well for the bacteria right it perceived a a perceived alpha it updated a it's all good so the following numerical example will demonstrate that minimizing free energy as it can lead to the exact opposite death this is really key fep haters out there note that the scope of the following numerical example is deliberately limited the goal is to demonstrate minimizing free energy can lead to

maladaptive inference when performed with the wrong priors all things being kept fixed if the priors are allowed to update the inference should lead to adaptive behavior so it's a contrived example this isn't how you would do fep modeling if you wanted your system to work adaptivity is guaranteed to the extent which priors match environmental constraints more than by the machinery employed to perform the inference so just saying we used bays it's like this isn't a sales pitch you could have the wrong priors we used free energy minimization in our software okay well you're going right off a cliff so you can think have the prior this is fine so it's not enough to hide behind the methodology the priors have to be stated and they have to be adaptive it's not enough just to say well free energy minimization occurred so things worked right no it's not just like your one stop shop it's like a linear model but a little bit different and more complex the following example of free energy minimization is provided with the sole purpose of supporting our response to the entailment problem which is to say supporting the weak and refuting the strong claim the goal is to give a formal intuition as to why minimizing free energy is not sufficient for life understood as the preservation of structural integrity by no means should the following numerical example be viewed as an exemplar of the manner in which free energy minimization operates mathematically okay so this is not an fep model it's just not it's an fep like model that we're using to make some important claims about fep and so here is the um in equation four is the free energy formulation this is the variational inference which we're not gonna go into um right now but um what we'll note is just the $q(r)$ of r that r is remember the receptor state like α β corresponds to the proposal recognition or approximate posterior density referred to in the fpp literature it is $q(r)$ the recognition density remember we're only making a sensory model that is embodied by the organisms not to be confused with the $p(r)$ and α which would be the joint distribution or generative model so way back in live stream number six we talked about how there's like the recognition model which is from the data to the hyper parameters that's from the observation of α or β to a or b and that's related to perception the world is handing us α or β and then we're updating our causal model of the world but then we're not here discussing action it's kind of a limited example it's more like a unidirectional example but adding in the second direction doesn't get around the fact that perceptual free energy was minimized and that doesn't always work so let's look at it not work so we're going to it's like and it works because it doesn't so let's assume our organism operates under variational bays and still represented a with the same level of confidence so here's what happens this is actually a good variational base case okay so a is observed and the fe is calculated at 0.47 in six β is observed or b it's a question of

whether b has a higher free energy contingent on observing a right so that gives a free energy of 2.297 2792 so minimizing free energy is better so under the variational model we still would have selected the right outcome so this is actually the variational equation working we're going to change it in a second variational equation works person's like yeah keep going minimizing free energy leads to survival only under the right conditions which is if you have the right priors so now we're going to imagine the same scenario with inverted maladaptive priors okay so you're dumped into the desert but the priors are for the ocean or something so here the p of b is 0.8 so now after there's an update the energy free energy of representing b after sensing a is 0.96 and the free energy when representing a after sensing α which is like the correct thing to do is higher it's over one so the minimal free energy would be representing b after sensing α because you had such an overwhelming belief in b that even observing a doesn't push you all the way to believing a i think there's a typo in the paper because it says represented state b with 0.63 percent confidence but it's actually 63 percent or 0.63 right but the point is if you started at 80 believing in b then you got α you dropped from 0.8 to 0.63 so you're still 63 you're less confident but still you're two-thirds believing in b so in other words free energy minimization just happened and you just picked the wrong thing because your priors were wrong that bacteria is gonna die so that's the hand sweeping those maladaptive priors off the table that's like evolution by natural selection hence minimizing free energy per se does not entail life not under the wrong prior right not dress like that you don't but right it's just again making a simple important point minimizing free energy is not sufficient for being a life it is necessary for being alive although free energy minimization is not the numerical examples suggest there may be an entailment that goes the other way around if you're alive it might very well be because you did something like minimize free energy so that's the weak relationship so we kind of invalidated the strong claim sufficiency but we're actually finding support for the weak claim of necessity this makes the FEP an interesting epistemic principle for researchers interested in development of themselves of cities of organisms in a reverse engineering fashion if we observe a free energy minimizing organism still living at the time that we observe it just like all things that we observe are we can trust it has good enough priors to remain alive like i look at the ants i think whatever they're doing as the world has changed they've been doing it for 120 million years that's what we know about that oh well how do they know what to do when there's a break in the trail well the ones that didn't aren't here and the ones that are here did it well enough and that's like a first pass at an answer and then also on a wing and a prior on a wing um means like sort of like a few

different things but it reminded me of the winged snowflake the 2007 paper and then moth to a flame and so um if you're observing these snowflakes the ones you observe are going to increasingly be the ones that have wings that can stay solid because the ones that don't don't exist so that's fep on a wing and a prior yep one could rightfully say that minimizing free energy when equipped with the right prior beliefs is sufficient for life i.e all you need to be qualified is living or maintaining your structural integrity the point here is the choice of prior whether under bayesian or approximate bayesian regime is a real concern since the entailment relation between life and fep entirely depends on the adaptivity of those priors interestingly some have argued the adaptivity of priors can also be guaranteed by a free energy minimization operating at the population level as a form of natural selection the citation there was to hesp 2019 a multi-scale proposal of the emerging complexity of life um with hesp ramsay constant bad kirchhoff and fristen and here's a figure from there i'll say though that upcoming there will be a paper with myself axel blue par in fristen that will go into a lot more detail on this so it's absolutely a great point and i believe that it's the case dean what did you write here well i was just wondering um from the perspective of what a prior is like is it available there and in terms of something that we can leverage off of and then of what what then what do we do because we have some sort of previous marker from which to to infer eventually so i was wondering if that's if that's the essence of what an adaptive adaptive as function as an um requires but but i think i'll wait until we bring up that paper with five authors well yeah exactly we will discuss it um we'll have fun things to discuss i can think of more examples but i'll save them so in the future direction last section the dissolution of the entanglement problem you know again write a paper if you don't think that the entitlement problem is adequately parsed and dissolved in this paper it's a great paper if somebody wanted to write it puts us in a good position to move to another related difficulty in the philosophical literature on the fep which is this time of an exegetic kind look at these synonyms hermeneutical there's an awesome fristen paper about the hermeneutics of communication these are just nice adjectives if minimizing free energy is not sufficient for life or survival how should we interpret statements such as the minimization of free energy may be if not sufficient characteristic of evolutionary successful system systems fristen and steph in 2007 way way way back like saying that the hawks gene duplication allowed for this body plan to change it doesn't say about what will happen but it is about what was necessary for happening so a statement such as the minimization of free energy may be a necessary if not sufficient characteristic of evolutionary successful systems this quote is probably such a post-dictive

statement the statement should be interpreted as claiming that free energy minimization must have occurred if a system is evolutionarily successful not the other way around we're looking at living creatures and postdicting what got them there looking at the class of all systems hypothetical failing etc and then saying what will make them alive because you can again minimize your free burn yourself out like a candle that'll minimize free energy too you could have maladaptive priors and also be minimizing your free energy daniel do you think this talks about the difference between deductively realizing when and inductively realizing when or is that let's let's let's talk about um deduction induction and abduction i think that's the win the win piece of this again i don't want to push this too far into the future because we're almost at the end here but i'm just maybe that's a point one thing i'm not sure yeah it's a it's a definite point one the final slide it's also the final lines of the paper um it's a little bit of a personal you know it's interesting to read a paper with i it's a first off one author paper but it's also more of like a philosophy and dupri's papers are written in this way as well when they're single author it's like i argue that this so that's kind of an interesting style that um stem papers usually don't have this is axel kind of bringing it all together as a person who used the free energy principle in the numerical example above the puzzling trace for which i was seeking a causal explanation for was the survival of the organism that was my observation the causal story or hypothesis for that observation under the conditions we imposed to our simulated organism was the free energy principle the inference over which led me to write the paper you are reading at the moment and the live stream you're listening to at this moment that paper functioned as sensory evidence for my hypothesis eg when writing down the number and seeing they were adding up and that paper live stream is the observation that you are using to evidence your hypothesis concerning the claim i said at the start of the paper namely that free energy so if you had wildly out of whack priors on that claim maybe even listening to us talk you still believe what you believe that's okay it's all consistent and good fidel to the unifying grip of hypotheses in the historical sciences the free energy principle is meant to account for all of that so here is axel looking at his figure one and then we're reading the paper and then like active lab and the broader ecosystem is observing that and there's other things observing us so fep all i see yeah again back to the question of whether it can be both objective and subjective at the same time and when whether that fits within the rel relatively small time frames of what we're looking at when we're talking about fep and this is even a smaller version of fpp than maybe others looking in terms of the unit of analysis anyway i i just think it the way he was able to use math to show there that there is a way of arriving at knowing what is

the um numerator and what's the denominator here was pretty a a was clear b it was pretty concise given the complexity and see it's pretty compelling so agree with all of that really looking forward to the dot one and dot two of course axel and anyone else is welcome to join in those discussions and then we have just so many more papers to discuss in the coming years we're gonna just do what we do in the dot one and dot two we're gonna be in the bathtub phase be in the thick of it be in the forest for the dot one and then we're gonna have our our springboard in our jumping off in the dot two we looked forward to this dot zero but soon we're going to be looking back on it what we did yeah exactly perfect perfect setup okay thank you so much absolutely um thanks everybody for watching hope to see you around alright bye